

Group 6 Term Project: Agentic RAG for Business Strategy

Course: BNAN 570 Generative AI: Insights and Strategy **Weight:** 30% of Final Grade

Project Overview

This project proposes the development of an Agentic Retrieval-Augmented Generation (RAG) system to enhance pharmacovigilance risk assessment within pharmaceutical companies.

Pharmacovigilance is the business function responsible for monitoring drug safety after products are released to the market. Companies are required to continuously review adverse event reports, evaluate potential safety signals, consult medical literature, and submit structured risk assessments to regulatory agencies. This process is highly manual, document-heavy, and time-consuming. Delays in signal detection increase regulatory and litigation risk, while inconsistent evaluations reduce operational efficiency.

The proposed system will orchestrate multiple AI agents to support this workflow. When a safety query is submitted (e.g., “Assess whether Drug X shows evidence of liver toxicity”), the system will retrieve relevant safety reports, analyze frequency patterns, reference regulatory guidance, and generate a structured safety summary. The objective is to augment analysts, reduce research time, and standardize early-stage signal triage. Strategically, this system supports enterprise risk mitigation and regulatory readiness.

1. Data Strategy & Source Requirements

Real-World Databases

The primary structured data source will be the FDA Adverse Event Reporting System (FAERS), which contains public adverse event reports including drug name, reported reaction, severity, and outcomes. This dataset enables basic signal trend analysis.

Unstructured data sources will include open-access PubMed abstracts related to drug safety and FDA drug label documentation containing known warnings and side effects. These text-heavy documents will form the retrieval knowledge base.

Synthetic Data Generation

If FAERS proves computationally intensive for Colab, a synthetic adverse event dataset will be generated using a defined schema (Drug Name, Adverse Event, Severity, Age Group, Outcome, Report Date). LLMs will generate realistic but simulated reports to populate the vector store.

Data Characteristics

The system will integrate both structured and unstructured data. Structured data supports analytical tasks such as frequency comparisons, while unstructured documents require chunking and embedding for semantic retrieval. Text documents will be segmented into

meaningful sections and stored in ChromaDB using OpenAI embeddings. The hybrid data design supports specialized agent roles within the workflow.

2. Technical Stack

The project will be developed in Google Colab using Python. OpenAI APIs will be used for language modeling and embeddings. LangChain will orchestrate the RAG pipeline and multi-agent workflow. ChromaDB will serve as the vector store for embeddings. Gradio will provide the user interface for query interaction.

3. Phased Deliverables

- Phase 1: Project Proposal & Data Strategy

This document defines the pharmacovigilance signal triage workflow and outlines the data sources and required agents.

Planned agent roles include:

Research Retrieval Agent: Retrieves relevant literature and safety documents.

Adverse Event Analysis Agent: Evaluates structured event data for signal trends.

Regulatory Context Agent: Provides compliance and safety threshold references.

Synthesis Agent: Generates structured safety assessment summaries.

- Phase 2: Core RAG Pipeline Construction

Implement document loading, chunking, embedding, and storage in ChromaDB. Develop a basic Gradio interface to test retrieval performance and evaluate grounding quality.

- Phase 3: Agentic Orchestration & Refinement

Transition to a multi-agent workflow where specialized agents collaborate sequentially to retrieve evidence, analyze data, and generate structured reports. Evaluate multi-turn context handling.

- Phase 4: Final Presentation & System Delivery

Deliver a live demonstration, fully commented Colab notebook, and critical analysis of performance, governance requirements, and business transformation needed for production deployment.